

CC Energy Coefficients

Where the 1, the $\frac{1}{4}$ and the $\frac{1}{2}$ in the coupled cluster energy expression come from, what happens to them under other conventions, and how to check them by hand.

Written for a graduate student who has read the refresher

Spin orbitals, antisymmetrized integrals, unrestricted sums unless stated otherwise

NOTATION

i, j, k / a, b, c / p, q, r, s	occupied / virtual / general spin orbitals; $ 0\rangle$ the reference, $E_0 = \langle 0 H 0\rangle$.
$\langle pq rs\rangle$	antisymmetrized integral $\langle pq rs\rangle - \langle pq sr\rangle$; antisymmetric under $p \leftrightarrow q$ and under $r \leftrightarrow s$.
τ_i^a, τ_{ij}^{ab}	$a_a^\dagger a_i$ and $a_a^\dagger a_b^\dagger a_j a_i$; $\tau_{ij}^{ab} 0\rangle = \Phi_{ij}^{ab}\rangle$.
t_i^a, t_{ij}^{ab}	amplitudes; $t_{ij}^{ab} = -t_{ji}^{ab} = -t_{ij}^{ba}$.
$\sum_{ijab}$ versus $\sum_{i<j, a<b}$	unrestricted sum over all index values versus restricted sum over distinct excitations.

THE FORMULA UNDER DISCUSSION

$$E = E_0 + \sum_{ia} f_{ia} t_i^a + \frac{1}{4} \sum_{ijab} \langle ij || ab \rangle t_{ij}^{ab} + \frac{1}{2} \sum_{ijab} \langle ij || ab \rangle t_i^a t_j^b \quad (1)$$

- 1 The structure $E = \langle 0 | H(1 + T_1 + T_2 + \frac{1}{2}T_1^2) | 0 \rangle$ comes from the exponential; the $\frac{1}{2}$ is the $1/2!$ of its series.
- $\frac{1}{4}$ The $\frac{1}{4}$ is the normalization of T_2 written as an unrestricted sum over four equivalent index orderings. It is the same $\frac{1}{4}$ that stands in front of the two-electron part of H .
- 1 Every term is a Slater-rule matrix element, f_{ia} or $\langle ij || ab \rangle$, times the coefficient of the corresponding determinant in the wavefunction. The energy is $\langle 0 | H | \Psi \rangle$ and nothing more.

1 The three terms

The energy is the projection of the Schrödinger equation onto the reference,

$$E = \langle 0 | H e^T | 0 \rangle = \langle 0 | H \left(1 + T + \frac{1}{2}T^2 + \frac{1}{6}T^3 + \dots \right) | 0 \rangle. \quad (2)$$

A two-body Hamiltonian connects the reference only to the reference, singles and doubles:

$$\langle 0 | H | 0 \rangle = E_0, \quad \langle 0 | H | \Phi_i^a \rangle = f_{ia}, \quad \langle 0 | H | \Phi_{ij}^{ab} \rangle = \langle ij || ab \rangle, \quad \langle 0 | H | \Phi_{ijk}^{abc} \rangle = 0. \quad (3)$$

With $T = T_1 + T_2 + \dots$, the only pieces of (2) that produce excitations of rank two or less are T_1 , T_2 , and $\frac{1}{2}T_1^2$. Everything else, T_3 , T_1T_2 , $\frac{1}{2}T_2^2$, $\frac{1}{6}T_1^3$ and so on, lands on triples or higher and is annihilated by $\langle 0 | H$. Hence

KEY RESULT · THE STRUCTURE BEFORE ANY INTEGRALS

$$E = E_0 + \langle 0 | HT_1 | 0 \rangle + \langle 0 | HT_2 | 0 \rangle + \frac{1}{2} \langle 0 | HT_1^2 | 0 \rangle. \quad (4)$$

Three terms, with coefficients 1, 1 and $\frac{1}{2}$. The $\frac{1}{2}$ is the $1/2!$ of the exponential series and has no other origin. This structure is the same for CCSD, CCSDT and every higher level, since higher clusters cannot reach the reference through H directly; they influence the energy only by changing t_i^a and t_{ij}^{ab} .

2 Unrestricted sums and antisymmetry

The second source of numbers is the convention for writing operators with antisymmetric indices. The two-electron part of the Hamiltonian is

$$H_2 = \frac{1}{4} \sum_{pqrs} \langle pq || rs \rangle a_p^\dagger a_q^\dagger a_s a_r. \quad (5)$$

Swapping $p \leftrightarrow q$ changes the sign of the integral and the sign of the operator string, so the term is unchanged; the same holds for $r \leftrightarrow s$. The unrestricted sum therefore visits each physically distinct term four times, and the $\frac{1}{4}$ corrects the count. Written with non-antisymmetrized integrals the same operator reads $\frac{1}{2} \sum \langle pq | rs \rangle a_p^\dagger a_q^\dagger a_s a_r$, where only the simultaneous swap of both pairs is a symmetry.

The doubles cluster operator follows the identical convention,

$$T_2 = \frac{1}{4} \sum_{ijab} t_{ij}^{ab} \tau_{ij}^{ab} = \sum_{i < j, a < b} t_{ij}^{ab} \tau_{ij}^{ab}, \quad (6)$$

because t_{ij}^{ab} and τ_{ij}^{ab} are both antisymmetric in each index pair, so the orderings (ij, ab) , (ji, ab) , (ij, ba) , (ji, ba) are the same operator. The general rule: a rank- n object with n antisymmetric hole indices and n antisymmetric particle indices carries $1/(n!)^2$ in front of an unrestricted sum. That is 1 for singles, $\frac{1}{4}$ for doubles, and $1/36$ for triples, which is the prefactor of the (T) energy formula.

WHERE THE $\frac{1}{4}$ OF THE HAMILTONIAN GOES

The matrix element $\langle 0 | H | \Phi_{ij}^{ab} \rangle = \langle ij || ab \rangle$ in (3) has coefficient 1, not $\frac{1}{4}$, even though H_2 carries $\frac{1}{4}$. Evaluating it from (5), the only surviving terms are those with $\{p, q\} = \{i, j\}$ and $\{r, s\} = \{a, b\}$: four index orderings, each contributing $\langle ij || ab \rangle$ once the signs of the integral and of the operator string are taken into account. The $\frac{1}{4}$ of H is consumed by those four orderings inside the Slater rule. This is why the energy (1) contains exactly one $\frac{1}{4}$ in its doubles term, inherited from T_2 , and not $1/16$.

3 Each term in turn

The singles term: coefficient 1

$$\langle 0|HT_1|0\rangle = \sum_{ia} t_i^a \langle 0|H|\Phi_i^a\rangle = \sum_{ia} f_{ia} t_i^a. \quad (7)$$

A single excitation has one hole and one particle index and no symmetry to compensate, so each appears once and the coefficient is 1. The Slater rule gives $\langle 0|H|\Phi_i^a\rangle = h_{ia} + \sum_j \langle ij||aj\rangle = f_{ia}$. For a Hartree–Fock reference this vanishes by Brillouin’s theorem; for any other reference, such as DFT or ROHF orbitals, it is the first-order coupling of the singles to the energy.

The doubles term: coefficient $\frac{1}{4}$

$$\langle 0|HT_2|0\rangle = \frac{1}{4} \sum_{ijab} t_{ij}^{ab} \langle 0|H|\Phi_{ij}^{ab}\rangle = \frac{1}{4} \sum_{ijab} \langle ij||ab\rangle t_{ij}^{ab} = \sum_{i<j, a<b} \langle ij||ab\rangle t_{ij}^{ab}. \quad (8)$$

The $\frac{1}{4}$ is the one from (6), carried through unchanged. In the restricted form it is simply the integral times the amplitude, once per distinct double excitation.

The quadratic singles term: coefficient $\frac{1}{2}$

Here the $\frac{1}{2}$ is the $1/2!$ of the exponential. The work is in evaluating the operator. Write out the square as an unrestricted double sum over single excitations:

$$\frac{1}{2} \langle 0|HT_1^2|0\rangle = \frac{1}{2} \sum_{ia} \sum_{jb} t_i^a t_j^b \langle 0|H\tau_i^a \tau_j^b|0\rangle. \quad (9)$$

Two single excitations compose into a double. Anticommuting a_i past $a_b^\dagger$ and then a_i past a_j ,

$$\tau_i^a \tau_j^b = a_a^\dagger a_i a_b^\dagger a_j = -a_a^\dagger a_b^\dagger a_i a_j = a_a^\dagger a_b^\dagger a_j a_i = \tau_{ij}^{ab}, \quad (10)$$

for $i \neq j$ and $a \neq b$, and zero otherwise. So $\langle 0|H\tau_i^a \tau_j^b|0\rangle = \langle ij||ab\rangle$ by (3), which gives the last term of (1):

$$\frac{1}{2} \langle 0|HT_1^2|0\rangle = \frac{1}{2} \sum_{ijab} \langle ij||ab\rangle t_i^a t_j^b. \quad (11)$$

It is worth seeing how this unrestricted sum counts a given distinct determinant $|\Phi_{ij}^{ab}\rangle$ with $i < j$ and $a < b$. Four terms of (11) reach it:

ORDERING OF THE TWO SINGLES	TERM	EQUALS
$(i, a) (j, b)$	$t_i^a t_j^b \langle ij ab \rangle$	$+ t_i^a t_j^b \langle ij ab \rangle$
$(j, b) (i, a)$	$t_j^b t_i^a \langle ji ba \rangle$	$+ t_i^a t_j^b \langle ij ab \rangle$
$(i, b) (j, a)$	$t_i^b t_j^a \langle ij ba \rangle$	$- t_i^b t_j^a \langle ij ab \rangle$
$(j, a) (i, b)$	$t_j^a t_i^b \langle ji ab \rangle$	$- t_i^b t_j^a \langle ij ab \rangle$

Their sum is $2 \langle ij || ab \rangle (t_i^a t_j^b - t_i^b t_j^a)$, and the $\frac{1}{2}$ in front reduces it to one copy per distinct determinant:

$$\frac{1}{2} \sum_{ijab} \langle ij || ab \rangle t_i^a t_j^b = \sum_{i < j, a < b} \langle ij || ab \rangle (t_i^a t_j^b - t_i^b t_j^a) = \frac{1}{4} \sum_{ijab} \langle ij || ab \rangle (t_i^a t_j^b - t_i^b t_j^a). \quad (12)$$

The three forms are identical. The first is how (1) is usually printed; the third puts the quadratic term on the same footing as the T_2 term, with the antisymmetrized product of singles in the role of a doubles amplitude.

4 The energy as a projection of the wavefunction

Combining (8) and (12), the energy collapses to one statement:

KEY RESULT · THE ENERGY IS THE DOUBLES COEFFICIENT CONTRACTED WITH THE INTEGRALS

$$E = E_0 + \sum_{ia} f_{ia} c_i^a + \sum_{i < j, a < b} \langle ij || ab \rangle c_{ij}^{ab}, \quad c_i^a = t_i^a, \quad c_{ij}^{ab} = t_{ij}^{ab} + t_i^a t_j^b - t_i^b t_j^a. \quad (13)$$

Here c_i^a and c_{ij}^{ab} are the coefficients of the singly and doubly excited determinants in $e^T |0\rangle$. Equation (13) is just $E = \langle 0 | H | \Psi \rangle$ with the Slater rules applied, which is what the projected Schrödinger equation says it should be. The doubles coefficient is the CI-to-CC relation $C_2 = T_2 + \frac{1}{2} T_1^2$ written out in indices, and the combination $t_{ij}^{ab} + t_i^a t_j^b - t_i^b t_j^a$ is exactly the τ_{ij}^{ab} intermediate of the Stanton–Gauss working equations. A code that forms τ once and contracts it with $\langle ij || ab \rangle$ has computed the whole correlation energy.

The product form of the exponential from the notes on exponentials, $e^T = \prod_{\mu} (1 + t_{\mu} \tau_{\mu})$, gives the same coefficient by inspection: the determinant Φ_{ij}^{ab} is reached once by the double excitation $\mu = (ij \rightarrow ab)$, with amplitude t_{ij}^{ab} , and once by each of the two pairs of singles that compose into it, $(i \rightarrow a)(j \rightarrow b)$ with product $t_i^a t_j^b$ and $(i \rightarrow b)(j \rightarrow a)$ with product $t_i^b t_j^a$ and a minus sign from (10).

5 The same numbers from diagrams

Diagrammatic derivations never write unrestricted sums or exponential series; they draw each distinct contraction once and attach weight factors by rule. The two relevant rules are: a factor $1/2$ for every pair of *equivalent lines*, two lines of the same type that begin at the same vertex and end at the same vertex; and a factor $1/k!$ for k *equivalent vertices*, identical cluster vertices connected in the same way. Applied to the energy:

ENERGY DIAGRAM	EQUIVALENT LINES	EQUIVALENT VERTICES	WEIGHT
H joined to one T_1 vertex	none	none	1
H joined to one T_2 vertex	two hole lines, two particle lines: two pairs	none	$1/2 \cdot 1/2 = 1/4$
H joined to two T_1 vertices	none, the lines go to different vertices	the two T_1 vertices	$1/2! = 1/2$

The algebra of Sections 2 and 3 and the diagram rules are the same counting seen from two sides. The equivalent-line factor is the $1/(n!)^2$ of the unrestricted sum, and the equivalent-vertex factor is the $1/k!$ of the exponential series. When the two bookkeepings disagree in a derivation, one of them has counted an ordering twice.

6 Other conventions

The coefficients in (1) belong to one convention: spin orbitals, antisymmetrized integrals, and unrestricted sums. Change the convention and the numbers change with it.

- **Closed-shell spatial orbitals.** Summing over spin and using spatial amplitudes t_{ij}^{ab} that are no longer antisymmetric gives

$$E_{\text{corr}} = \sum_{ijab} [2(ia|jb) - (ib|ja)] (t_{ij}^{ab} + t_i^a t_j^b), \quad (14)$$

with chemists' integrals, no $\frac{1}{4}$ and no $\frac{1}{2}$, and the exchange product $t_i^b t_j^a$ gone from the parenthesis. The 2 and the -1 are what remains of spin integration. This is the form most closed-shell codes implement.

- **Restricted sums.** Writing every sum over $i < j, a < b$ removes the $\frac{1}{4}$ and turns the $\frac{1}{2}$ into the antisymmetrized product, as in (13). Some authors print the energy this way, and the $\frac{1}{4}$ and $\frac{1}{2}$ then never appear.
- **Mixed electron–proton terms.** In multicomponent CCSD the electron–proton contribution to the energy is $-\sum_{iIaA} (ia|IA) (t_{iI}^{aA} + t_i^a t_I^A)$, with coefficient 1. An electronic index and a protonic index cannot be interchanged, so there is only one ordering per term and no symmetry factor to compensate.
- **Perturbative triples.** The $1/36$ in the (T) energy formula is $1/(3!)^2$, the rank-three case of the rule in Section 2.

If a coefficient in a paper or a code looks wrong, the first thing to check is which of these conventions is in force. Most apparent discrepancies are a $\frac{1}{4}$ that has been absorbed into a restricted sum or a 2 that has come from spin integration.

7 A check by hand

Take the two-electron, two-orbital system of the refresher: occupied spin orbitals $g\alpha, g\beta$, virtual spin orbitals $u\alpha, u\beta$, and a single doubles amplitude $t = t_{g\alpha g\beta}^{u\alpha u\beta}$. The exact correlation energy is Kt , with $K = \langle g\alpha g\beta || u\alpha u\beta \rangle$. The unrestricted sum in (8) runs over sixteen index combinations, of which four are nonzero:

(I, J, A, B)	$\langle IJ AB \rangle$	T_{IJ}^{AB}	PRODUCT
$(g\alpha, g\beta, u\alpha, u\beta)$	$+K$	$+t$	$+Kt$
$(g\beta, g\alpha, u\alpha, u\beta)$	$-K$	$-t$	$+Kt$
$(g\alpha, g\beta, u\beta, u\alpha)$	$-K$	$-t$	$+Kt$
$(g\beta, g\alpha, u\beta, u\alpha)$	$+K$	$+t$	$+Kt$

The sum is $4Kt$, and the $\frac{1}{4}$ brings it back to Kt . The same exercise with the T_1^2 term, for a system where singles are allowed, reproduces the table in Section 3. Doing this once, with real indices,

is the fastest way to make the coefficients stop looking arbitrary.

8 Further reading

1. A. Szabo and N. S. Ostlund, *Modern Quantum Chemistry* (Dover, 1996), Chapter 2. The Slater rules in first and second quantization, and the $\frac{1}{4}$ and $\frac{1}{2}$ conventions for the Hamiltonian.
2. T. D. Crawford and H. F. Schaefer III, “An introduction to coupled cluster theory for computational chemists”, *Rev. Comput. Chem.* **14**, 33 (2000). The energy derived algebraically and diagrammatically, with the weight rules.
3. I. Shavitt and R. J. Bartlett, *Many-Body Methods in Chemistry and Physics* (Cambridge, 2009), Chapters 4 and 9. The diagram rules for equivalent lines and vertices, and the unrestricted-sum conventions.
4. J. F. Stanton, J. Gauss, J. D. Watts and R. J. Bartlett, *J. Chem. Phys.* **94**, 4334 (1991). The τ intermediate and the working equations in which it appears.

Equations are numbered consecutively through the notes. The closed-shell form (14) uses spatial orbitals and chemists' notation for the integrals.